



पदार्थ विज्ञान एवं अभियांत्रिकी विभाग

भारतीय प्रौद्योगिकी संस्थान कानपुर

Department of Materials Science and Engineering  
Indian Institute of Technology Kanpur

MATERIALS  
PEOPLE  
SOLUTIONS  
A BRIGHTER  
TOMORROW



# PLACEMENT BROCHURE

2026-27

ENGINEERING MATERIALS  
FOR A SUSTAINABLE TOMORROW



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**IIT KANPUR**  
Department of Materials  
Science & Engineering

*Your journey to impact, innovation and leadership starts here.*



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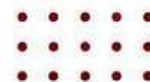


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## HEAD OF THE DEPARTMENT'S OPEN LETTER



### Dear Recruiter,

Materials Science and Engineering (MSE) department presents students, who are well-equipped with knowledge, skills and values essential for excelling in ever-evolving professional landscape.

MSE department fosters an environment of excellence, commitment and holistic development. We strongly believe in equipping our students with not only theoretical knowledge but also exposing practical/laboratory skills that are crucial in today's dynamic and competitive world.

MSE students are well-prepared to tackle real-world challenges undergoing rigorous curriculum, hands-on projects, internships, and industry interactions, so as to meaningfully contribute to your esteemed organizations. We take pride in the fact that MSE students excel in wide spectrum and are eager to step into industries, like automotive, aerospace, materials processing, iron and steel making, non-ferrous metallurgy, ceramics, healthcare, semiconductors, data analytics including AI/ML, and many more.

While 35% of our undergraduates of our department are trained to acquire basic understanding of the materials and quantitative analysis so that they can take up a wide variety of problem-solving challenges offered by the recruiter, our research cadre students, 70 M.Tech students, 06 BT-MT students and 100 PhDs, are provided specialized training to take up any scientific and technical challenges for the development and utilization of new processes and materials for key projects in your futuristic strive to become global leaders. With a strong foundation in mathematical and numerical methods, MSE students are also eager and capable of contributing to building softwares for the fulfilment and knowledge creation in materials world. Beyond academics, we emphasize the importance of soft skills, leadership qualities, civic responsibility, integrity and ethical values. Our students participate in various extracurricular activities, community service initiatives, and industry interactions, which contribute to their overall personality development and prepare them to be responsible global citizens.

By recruiting from Department of Materials Science and Engineering at IIT Kanpur, you will be investing in individuals who are not only technically proficient but also possess the drive and determination to make meaningful contributions to your organization. We sincerely wish that some of our students will have the privilege of becoming part of your esteemed organization. Be assured that MSE students at IIT Kanpur are nothing short of being remarkable, adding substantial value to your esteemed institution.

MSE department wishes to build a strong relation through this recruitment and extends a sincere 'thank you' for considering our students for various opportunities within your honoured organization. We look forward to establishing a mutually beneficial partnership and contributing constructively to your admired success.



**Prof. Kantesh Balani**

Head, Materials Science and Engineering  
IIT Kanpur



# About Us



The **Department of Materials Science and Engineering** at IIT Kanpur with its 60 long years of legacy of excellence in undergraduate and post graduate teaching and research strives to prepare technologists and engineers to develop new materials and processes for applications in industries of metal and mining, automotive, chemical, aviation, plastic, biotechnology, semiconductor, solar and energy sectors.

The Department was established in 1960. Founded in 1960, the **Department of Materials Science & Engineering** at IIT Kanpur holds a prominent position among the nation's leading schools in Materials Science and Engineering. Initially named **Metallurgical Engineering**, the department introduced its postgraduate program in 1964. Over the years, it underwent a series of adaptations as per the need of the country and society, changing from **Metallurgical Engineering** to **Materials and Metallurgical Engineering** in 1993 and eventually transitioning to the Department of Materials Science and Engineering in 2009. Since its inception, it has had a strong impact in providing knowledgeable manpower to meet the nation's demand in traditional metallurgy. The department consciously nurtures overall professional growth of the students, apart from giving quality education. The department has constantly reinvented to keep the IIT curriculum in pace with the state-of-the-art technologies.

The field of study in the department now encompasses the entire spectrum of **extractive metallurgy, physical metallurgy, manufacturing processes, electronic and semiconductor materials, mechanical behavior of materials, powder metallurgy, process modeling, material degradation, nanomaterials, biomaterials, ceramics, composites, recycling & material recovery and computational materials engineering (AI/ML), microstructure, materials processing, physical & mechanical metallurgy, coatings & materials for damage tolerance (for fatigue, creep, fracture, etc.** This department has pioneered a unified approach for teaching and research, which has enabled us to evolve into an interdisciplinary field contributing to diverse applications and technological development.





# PLACEMENT TIMELINE

## 2026-27

### KEY MILESTONES IN THE RECRUITMENT PROCESS



“We look forward to your continued support in shaping the future with us.”



## ACADEMIC PROGRAMS

**B. Tech**  
400

**BT-MT**  
06

**M. Tech**  
70

**PhD**  
100



MSE Department at IIT Kanpur puts emphasis on enhancing the technical expertise of the students and provides hands-on Industrial exposure to students with internships, projects, and visits.

### UG Coursework

- ▶ Basic Sciences
- ▶ Core Laboratories
- ▶ Departmental Courses
- ▶ Internship (2nd /3rd year)
- ▶ B.Tech Thesis
- ▶ Internship



### PG Coursework

- ▶ Structure and Characterization of Materials
- ▶ Transport Phenomena
- ▶ Thermodynamics
- ▶ Mathematical and Computational Methods
- ▶ M.Tech/Ph.D Thesis
- ▶ Teaching Assistantship
- ▶ Internship





# RESEARCH FACILITIES

MSE department at IIT Kanpur is furnished with world-class research facilities assisting students in learning vital skills and gain hands-on experience of the latest technologies used in industries and academia. Apart from the various testing and characterization laboratories, the department also houses befitting computational and modeling facilities in steelmaking, fluid dynamics, and solidification processing. We have the following labs in our department: <https://iitk.ac.in/mse/MSE-Facilities/>



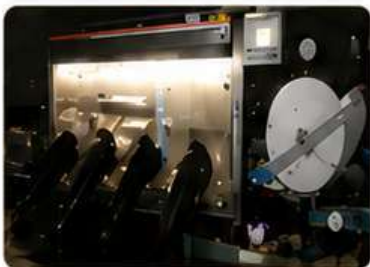
## MICROSTRUCTURE CHARACTERIZATION FACILITY

- Scanning Electron Microscopy (SEM), Atomic Force Microscopy (AFM)
- Transmission Electron Microscopy (TEM), Bruker-Emmet-Teller (BET)
- Electron Probe Micro Analyzer (EPMA), Thermogravimetric Analysis (TGA) and Differential Scanning Calorimetry (DSC), X-Ray Diffraction (XRD), X-Ray Photoelectron Spectroscopy (XPS)



## PHYSICAL METALLURGY AND ENGINEERING METALLURGY LAB

- Optical Microscopy
- Microwave Sintering Furnace
- Rolling Mill
- Welding
- Brazing



## ELECTRONIC MATERIALS AND THIN FILM PROCESSING LAB

- Pulse Laser Deposition
- Electron-Beam Evaporation
- Photolithography
- Clean Room
- Sputtering



## MATERIAL TESTING LAB

- Universal Testing Machine
- Fatigue Testing
- Creep Testing
- Impact Testing
- Hardness Testing
- Microhardness Testing





# RESEARCH CENTRES



## SAMTEL CENTRE FOR DISPLAY TECHNOLOGIES

To conduct R&D so as to nurture and support the growth of science and technology of electronic displays and to establish a tripartite relationship between industry, academia and governmental agencies.



## NATIONAL CENTRE FOR FLEXIBLE ELECTRONICS

It acts as a nodal point in India to bring academia, industry and public research organizations under one umbrella for research and development of large area flexible electronics.

**INDUSTRY PARTNERS:** Applied Materials, Manipal Technologies, Chain Electronics, Mathura Manufacturing



## ADVANCED CENTRE FOR MATERIALS SCIENCE

Advanced Centre for Materials Science was created in 1978 with a view to make available major materials preparation and characterization facilities under one roof. These state-of-the-art research facilities are regularly upgraded, and maintained by suitably trained competent staff.



**ICME** NATIONAL HUB  
AN IITK-TCS JOINT INITIATIVE

## INTEGRATED COMPUTATIONAL MATERIALS ENGINEERING

Integrated Computational Materials Engineering is a National Hub at IIT Kanpur – A Joint IITK-TCS Initiative.



## CENTRE FOR NANOSCIENCES

The centre is aimed to provide various nanomission & nanotechnology related fabrication & characterization tools for fundamental research support to startups and other academic & industrial partners.





## Ongoing Projects

- OLEDs for short range optical transmitters and visible light communication (VLC).
- E-Beam assisted Thin Film Fabrication of Binary Metal Oxide RRAM Device for resistive switching.
- Study of Resistive Switching Behavior in Sulfur-Based Materials for RRAM Applications.
- Fabrication and Development of a Flexible Piezoelectric Touch-Integrated OLED Display.
- Fabrication of Antimony Sulfo-selenide Thin Films for Photoelectrochemical Applications.
- Interfacial Engineering of Mg-Doped NZSP Solid Electrolytes Using Inorganic Interlayers for Stable Na metal Plating/Stripping.
- Anisotropy-Driven Evolution of Skyrmion Domain Walls: From Easy-Axis to Easy-Plane Regimes.
- Nanocomposite coatings for anticorrosive, antimicrobial, or antifouling properties.
- Hydrogen assisted deformation in high-manganese (Mn) steel.
- Recycling of aluminium scrap through hydrometallurgy.
- Density Functional Theory and AI/ML Based Exploration Of Composition-Structure-Property Relationship in High-Entropy Shape Memory Alloys.
- Mesoscale Modeling of Anisotropic Hydrogen Diffusion and Diffusion-Induced Stress Generation in Dual-Phase Titanium Alloys.
- Effect of Build Direction and Post-processing treatments on Creep behaviour of Selective Laser Melting based (Additive Manufacturing) AlSi10Mg alloy.
- Effect of Alloying Elements on the Phase-Specific Mechanical Properties of Eutectic Sn-Bi Solder Alloy Using Nanoindentation.
- Development of porous copper-based interconnects for fine pitch PCB assembly.
- Design and fabrication of a Thermoelectric Generator Device for Waste Heat Recovery.
- Arjun Tree Bark as a green sustainable corrosion inhibitor against chloride attack on Mild Steel.
- Electrochemical investigation of leaf extract as green corrosion inhibitor for High-Chromium Steel.
- Experimental and Simulation-Based Investigation of the Effect of DC Current on the Tensile Creep of Pure Aluminium and Al 6061-T6 Alloy.
- Engineering Electrode-Electrolyte Interfaces and High Mass Loading Architectures for Practical All-Solid-State Sodium Ion Batteries.



- Crystallization Characteristics of Mold Fluxes in Continuous Casting of Ti-bearing Steels.
- H<sub>2</sub> based sustainable iron making.
- Wear resistant High Entropy Alloy with dispersoids.
- Recycling of mild steel.
- Grain Boundary Engineering.
- Effect of adding H<sub>2</sub> and O<sub>2</sub> on plasma etching of Si and SiO<sub>2</sub> using various fluorocarbons and hydrofluorocarbons for a certain 3D structure using simulation
- Experimental characterization and multiscale scale-bridging simulation of deformation and microstructure–mechanical behavior in advanced multiphase steel by using DIC, Abaqus, and DAMASK crystal plasticity”.
- Additive Manufacturing of 17-4 PH Stainless Steel: Microstructural Evolution and Mechanical Properties.
- Hydrogen Embrittlement in AHSS (3rd Generation steel).
- Near-Infrared Light-Responsive Amphiphilic Block Copolymers for Controlled Drug Delivery and Photothermal Cancer Therapy.
- Sintering Behaviour of Nb with Ni-Cu Additives.
- Nanophotonic Design, Spectral Optimization, and Emissivity-Limit Analysis of Multilayer Distributed Bragg Reflectors for Passive Daytime Radiative Cooling in space applications.

## SOFTWARE & COMPUTATIONAL TOOLS

| ATOMIC & MOLECULAR SIMULATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | PROGRAMMING & DATA ANALYSIS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | MULTIPHYSICS & SIMULATION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>LAMMPS<br>Large-scale Atomic/Molecular<br>Massively Parallel Simulator<br><br><br>OVITO<br><br><br>VASP<br><br><br>VASPKit<br><br><br>Phonopy<br><br><br>pymatgen | <br>Python<br><br><br>NumPy<br><br><br>pandas<br><br><br>scikit-learn<br><br><br>XGBoost<br><br><br>Matplotlib<br><br><br>Seaborn<br><br><br>OpenCV<br><br><br>Linux / Ubuntu<br><br><br>Jupyter<br><br><br>Anaconda<br><br><br>MTEK<br>MTEK | <br>COMSOL<br>COMSOL Multiphysics<br><br><br>Ansys<br><br><br>Abaqus<br><br><br>NABINSA<br><br><br>CFD<br><br><br>Aspen Plus<br><br><br>SolidWorks<br><br><br>AutoCAD<br><br><br>LaTeX<br>LaTeX |
| MATERIALS CHARACTERIZATION & ANALYSIS                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | MACHINE LEARNING & AI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | OPTIMIZATION & MODELLING                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <br>VESTA<br><br><br>FTR Software<br><br><br>Origin<br><br><br>EZView<br><br><br>GSAS-II<br><br><br>Thermo-Calc                                                       | <br>TensorFlow<br><br><br>Keras<br><br><br>PyTorch<br><br><br>PCA<br><br><br>Regression<br><br><br>Random Forest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <br>Matlab<br>(Optimization)<br><br><br>FEM<br>Toolkit<br><br><br>Optimized<br>Solutions<br><br><br>L-OPTIM<br><br><br>Surrogate<br>Modeling<br><br><br>Inverse Design                                                                                                                                                                                                                                                                                         |

# Academic Curriculum

## Metallurgical Engineering



- Y Iron and Steel Making
- Y Principles of Metal Extraction and Refining
- Y Phase Transformations
- Y Thermodynamics & Phase Equilibria
- Y Mechanical Behaviour of Materials
- Y Materials Failure: Analysis and Prevention
- Y Structure and Characterization of Materials
- Y Diffusion in Solids
- Y Materials Recycling

## Nanomaterials and Biomaterials

- Y Introduction to Biomaterials
- Y Materials Science Technologies for Applications in Life Sciences
- Y Transmission Electron Microscope Nano Analysis of Materials
- Y Nanostructures and Nanomaterials
- Y Characterization and Properties
- Y Coatings



## Electronic Materials



- Y Electronic Devices and Characterization
- Y Electro-ceramic Materials and Applications
- Y Computer Simulations in Materials Science
- Y Technology of Thin Films and Device Fabrication
- Y Introduction to Semiconductor Packaging
- Y Energy Materials and Technologies
- Y Materials for Semiconductor Industry
- Y Display Technology

## Materials in Manufacturing

- Y Materials Processing
- Y Selection & Design of Engineering Materials
- Y Manufacturing Processes
- Y Solidification Processing
- Y Heat Treatment and Surface Hardening
- Y Powder Metallurgy
- Y Introduction to Lightweight Alloys





# Faculty List

**Nilesh Badwe**

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**Soumya Sridar**

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**Raghupathy Yuvraj**

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**Anish Upadhyaya**

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**Vivek Verma**

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**Shivam Tripathi**

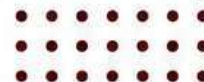
Website: <http://home.iitk.ac.in/~shivamt>

**Dipak Mazumdar**

*(Emeritus Faculty)*

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# Distinguished Alumni



**Mr. Suresh Pandey**  
(BT/MME/1965)  
Former Director,  
Bokaro Steel Plant  
(Management excellence)



**Prof. Raj N. Singh**  
(BT/MME/1967)  
Regents Professor,  
Oklahoma State University  
(Member of the National  
Academy of Engineering)



**Prof. Jagdish Narayan**  
(BT/MME/1969)  
Prof., Carolina  
State University  
(Academic Excellence)



**Mr. B. K. Shah**  
(BT/MME/1974)  
Exec. Director, AIA  
(Entrepreneurial  
Excellence)



**Mr. Som Mittal**  
(BT/MME/1973)  
Former Chairman,  
NASSCOM  
(Management Excellence)



**Mr. Anil Bansal**  
(BT/MME/1977)  
President, Realty National  
Management, Inc.  
(Distinguished  
Alumnus Award 2023)



**Mr. Jai Shankar Sharma**  
(BT/MME/1977)  
Mentor of the Bangalore  
Chapter, IIT Kanpur  
(Distinguished  
Alumnus Award 2023)



**Shree Pradeep Goyal**  
(BT/MME/1978)  
Founder Chairman and  
Managing Director of  
Pradeep Metals Limited,  
Mumbai)



**Prof. Veena Sahajwalla**  
(BT/MME/1986)  
Scientist Professor,  
UNSW  
(Academic Excellence)



**Dr. Pramath Raj Sinha**  
(BT/MME/1986)  
Founder,  
Ashoka University  
(Service of the society  
at large)



**Prof. Arvind Agarwal**  
(BT/MT/MME/1993/1995)  
(Professor, Florida  
International  
University (FIU), USA)



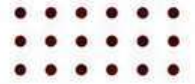
**Prof. Aparna Singh**  
(BT/MME/2007)  
Professor IIT Bombay  
(Young Metallurgist  
of the year)





## Past Recruiters





# Past Recruiters



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